

Shyam Krishna Khadka, PhD (soon-to-finish)

Network Security Engineer | Internet Security Researcher | Enterprise Network Architect

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[GitHub](#)

Professional Summary

Network Security Engineer with 15+ years of experience designing, deploying, and securing large-scale enterprise and service provider networks, complemented by cutting-edge research in Internet security. Currently completing a Ph.D. at University of Twente, Netherlands, specializing in BGP security, Internet measurement, and DDoS mitigation.

Core Expertise

- Networking: BGP, OSPF, MPLS, Routing, Switching, SD-WAN, WiFi6, CCNP training
- Security: NGFW, XDR, Firewalls
- Infrastructure: VMware, HCI, Linux, RHCE training
- Programming: Python, Java, Ruby, C/C++, JavaScript
- Systems: Linux, Git, Virtualization
- Research and leadership: Internet measurements, data analysis, technical lead, presentations

Professional Experience

Doctoral Researcher, University of Twente (2023–Present), The Netherlands

- Designed Internet-scale measurement methodologies related to the security of BGP and Autonomous Systems.
- Designed and implemented novel methodologies to detect exposed BGP routers and characterize global DDoS scrubbing deployments.
- Published and presented multiple peer-reviewed papers at leading global networking conferences.
- Present research findings at international networking forums, including RIPE, and IETF.
- Assisted in course Advanced Networking, mainly data-plane programming using P4.

Senior Engineer, Nepal Telecom (2010 – 2022)— Kathmandu, Nepal

Progressed from Software Developer and Network/System Engineer to Senior Engineer over nearly thirteen years in Nepal's largest telecommunications provider.

- Led technical evaluation and deployment of SD-WAN, HCI, EDR, Wi-Fi 6, next generation firewalls and enterprise networking projects that include multiple vendors such as Cisco, Juniper, CheckPoint, PaloAlto, VMware, and Nutanix.
- Managed day-to-day routing, switching, and Linux system administration.
- Handled technical procurement, compliance testing, and network operations.

- Developed internal web applications and services utilizing Java, ASP.NET, JavaScript, and Nginx.

Software Developer, (2009 – 2010), CloudFactory — Kathmandu, Nepal

- Developed web applications using Ruby on Rails.
- Built database-driven applications.
- Worked in Agile software development teams.
- Collaborated with international engineering teams.

Education

University of Twente — PhD Candidate (Expected 2026)

Researching Internet-scale network security with emphasis on routing security and resilient Internet infrastructure.

University of Crete (2016)— MSc Computer Science

Tribhuvan University (2015)— MSc Computer Engineering

Tribhuvan University (2009) — Bachelor of Computer Engineering

Tribhuvan University (2022)— Bachelor of Law (LLB)

Selected Publications

- Detecting and Characterizing Exposed BGP Routers (TMA 2026)
- Detecting and Characterizing DDoS Scrubbing from Global BGP Routing (PAM 2026)
- A First Look at the Adoption of BGP-based DDoS Scrubbing Services (CNSM 2025)
- Assessing the Security of Internet Paths (ANRW 2024)

Awards & Recognition

- IRTF Diversity Grant (IETF 126, Vienna, 2026)
- One of the three best measurement blogs for 2025 by APNIC
- Erasmus Mundus Scholarship at University of Crete
- Faculty Topper, Bachelor of Computer Engineering
- Full Scholarship, both Bachelor and MSc Computer Engineering

Languages

English – Professional proficiency, Nepali – Native, Dutch – Basic (A2)